

An Internship Report

Of

Web Development Training & Internship

At

TRL FutureX

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)

By



Roll Number : 2410030839

Name of student : Mahak Verma

Batch : 2024 -2028

CANDIDATE'S DECLARATION

I, **MAHAK VERMA** do hereby declare that the internship report titled “**Web Development Training & Internship at TRL FutureX**” has been completed by me to fulfil the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**.

All the references used in this report have been appropriately acknowledged. I have not knowingly taken any material from any other source without proper acknowledgement. I affirm that this report represents my own work and has not been submitted to any other institute or university for the fulfilment of any degree or diploma requirement. I further declare that I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: MAHAK VERMA

Roll No.: 2410030839

Batch: 2024–2028

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the **Internship Program**. I am thankful for their inspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on various aspects related to this internship.

I would like to express my sincere gratitude to **TRL FutureX** for providing me with the opportunity to participate in the **Web Development Training & Internship Program** and for providing a structured platform to learn and gain practical exposure in the field of **Web Development**. The internship helped me enhance my knowledge and practical skills in **HTML, CSS, JavaScript, responsive web design, and frontend development**.

I am also thankful to my institute, **IILM University, Greater Noida**, and the faculty of the **School of Computer Science and Engineering** for their continuous support, guidance, and encouragement throughout the internship.

I express my heartfelt gratitude to my **parents** for their blessings, constant support, and encouragement.

Signature: _____

Student Name: MAHAK VERMA

Roll No.: 2410030839

TABLE OF CONTENTS

1. Candidate's Declaration	i
2. Acknowledgement	ii
3. Internship Completion Certificate	iii
4. Project Description	1
4.1 Introduction	1
4.2 Organization Profile	2
4.3 Problem Statement	3
4.4 Project Objectives	3
4.5 Scope of the Project	4
4.6 Technologies and Tools Used	4
4.7 System Architecture	5
4.8 Methodology	6
4.9 Expected Outcomes	7
4.10 Certificates of Completion and Communication Proof	8
5. Bibliography/References	9

INTERNSHIP OFFER LETTER

INTERNSHIP OFFER LETTER

Dear Mahak Verma,

Date: 30/07/2026

We are pleased to extend to you an offer for the position of **Programmer Analyst Intern** at **TRL Futurex**. We believe that your skills and enthusiasm will make a valuable contribution to our team. This letter outlines the details of your internship with us.

Internship Details:

- **Designation:** Programmer Analyst Intern
- **Internship:** Web Development Training & Internship
- **Duration:** 1 Month
- **Start Date:** 30/07/2026
- **End Date:** 30/08/2026
- **Location:** Virtual/Physical Internship
- **Work Hours:** Monday to Friday, 9:30 AM - 6:30 PM

Roles and Responsibilities:

As a Programmer Analyst Intern, you will be expected to:

- Assist in developing and maintaining responsive websites and web applications.
- Write clean, efficient, and well-documented code.
- Collaborate with senior developers to implement project requirements.
- Debug, test, and optimize website performance.
- Integrate APIs and database functionality where required.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the **one-month Web Development Training & Internship** undertaken at **TRL FutureX** as part of the academic requirements of the **Bachelor of Technology (Computer Science and Engineering)** programme at **IILM University, Greater Noida**.

The primary aim of this internship was to develop a strong practical foundation in **Web Development** and to apply web technologies through hands-on assignments and projects. The internship provided practical exposure to **HTML, CSS, JavaScript, responsive web design, frontend development, and website implementation**.

The internship followed a structured, project-based approach in which different web development concepts were introduced progressively. The learning process started with basic webpage structure using HTML, followed by styling and responsive design using CSS, and then progressed to interactive web development using JavaScript. The final stage involved combining these skills to develop a complete **Smart Restaurant Booking Website**.

Through these activities, I gained practical experience in designing webpages, creating user interfaces, implementing interactive features, making websites responsive, and developing complete frontend projects.

4.2 Organization Profile

TRL FutureX provided the platform for the **Web Development Training & Internship Program**. The internship was structured to provide students with practical exposure to web development through learning modules, assignments, and project-based activities.

The training focused on developing fundamental skills required to create modern and responsive websites. It provided an opportunity to apply theoretical concepts through practical implementation rather than limiting the learning process to classroom-based theory.

During the internship, the major areas covered included:

- HTML and webpage structure
- CSS and webpage styling
- Responsive web design
- JavaScript programming
- DOM manipulation
- Interactive frontend development
- Form development
- User interface design
- Website development and testing

The project-oriented nature of the internship helped me understand how different web technologies work together to create functional and user-friendly websites.

4.3 Problem Statement

Web development requires more than theoretical knowledge of programming languages and markup technologies. Students often understand individual concepts such as HTML, CSS, and JavaScript but have limited practical experience in combining them to develop complete and responsive web applications.

The problem addressed during this internship was to gain practical experience in the complete frontend web development workflow — starting from creating structured webpages, applying CSS-based designs, developing responsive layouts, adding JavaScript-based interactivity, and finally integrating these skills into a complete web project.

The internship therefore focused on developing practical skills through a series of progressively challenging tasks and projects.

4.4 Project Objectives

The major objectives of the internship were:

- To understand the fundamentals of web development.
- To learn how to structure webpages using **HTML5**.
- To develop webpage designs using **CSS3**.
- To understand responsive web design principles.
- To learn how to create user-friendly web interfaces.
- To understand JavaScript fundamentals.
- To implement interactive features using JavaScript.
- To learn DOM manipulation and event handling.
- To develop functional web-based projects.
- To understand the integration of HTML, CSS, and JavaScript.
- To improve problem-solving and debugging skills.
- To develop a complete responsive website as a final project.
- To gain practical experience in frontend development.

4.5 Scope of the Project

The scope of this internship covered the fundamental and practical aspects of **frontend web development**.

The internship began with the development of a **Student Registration Webpage**, which provided practical experience in HTML structure and forms. The second module focused on CSS and responsive design through the development of a **Responsive Restaurant Website Homepage**.

The third module introduced JavaScript and interactive frontend development through a **To-Do List Web Application**. The final module combined the knowledge gained from the previous modules to develop a **Smart Restaurant Booking Website**.

The internship covered:

- Static webpage development

- HTML-based forms
- CSS styling
- Responsive layouts
- JavaScript functionality
- DOM manipulation
- Interactive user interfaces
- Frontend project development
- Website testing and debugging

The internship was conducted in an online training environment and focused primarily on **frontend web development**.

4.6 Technologies and Tools Used

The following technologies and tools were used during the internship:

Web Technologies

- **HTML5** – Used to create the structure and content of webpages.
- **CSS3** – Used for styling, layouts, spacing, typography, and responsive design.
- **JavaScript** – Used to implement interactive and dynamic functionality.

Web Development Concepts

- Responsive Web Design
- CSS Flexbox
- HTML Forms
- DOM Manipulation
- Event Handling
- User Interface Design
- Frontend Development

Development Tools

- Code Editor
- Web Browser
- Browser Developer Tools

These technologies and tools were used together to design, implement, test, and improve the projects completed during the internship.

4.7 System Architecture

The general architecture followed during the internship can be represented through the following layers:

1. Structure Layer

HTML5 forms the basic structure of the website. It defines elements such as headings, paragraphs, forms, buttons, navigation sections, images, and other webpage components.

2. Presentation Layer

CSS3 is used to control the visual appearance of the webpage. It manages colors, fonts, spacing, layouts, buttons, navigation elements, and responsive behavior.

3. Interaction Layer

JavaScript provides dynamic functionality and allows the webpage to respond to user actions.

4. DOM Layer

JavaScript interacts with the **Document Object Model (DOM)** to dynamically create, modify, or remove webpage elements.

5. Responsive Layer

Responsive design techniques ensure that the websites adapt to different screen sizes such as desktops, tablets, and mobile devices.

6. Application Layer

The different technologies are integrated to create functional applications such as the **To-Do List Web Application** and the **Smart Restaurant Booking Website**.

7. Testing Layer

The developed webpages and applications are tested using web browsers and developer tools to identify layout, functionality, and coding issues.

Suggested figure for this section:

User → HTML Structure → CSS Design → JavaScript Interaction → Web Application → Browser

4.8 Methodology

The internship followed a **module-based and project-oriented methodology**. Each module introduced new concepts and provided an opportunity to apply them through practical implementation.

Module	Project/Task	Description
1	Student Registration Webpage	Created a structured webpage using HTML with forms and different input elements for collecting student information.
2	Responsive Restaurant Website	Designed a visually appealing restaurant homepage using CSS and implemented responsive layouts for different screen sizes.
3	To-Do List Web Application	Developed an interactive task-management application using JavaScript, DOM manipulation, and event handling.

4	Smart Restaurant Booking Website	Combined HTML, CSS, and JavaScript to develop a complete responsive restaurant website with booking-related functionality.
---	---	--

Module 1 – HTML & Web Foundations

The first module focused on understanding HTML and creating the structure of webpages. A **Student Registration Webpage** was developed using forms, input fields, labels, selection controls, and buttons.

Module 2 – CSS & Responsive Web Design

The second module focused on designing webpages using CSS. A **Responsive Restaurant Website Homepage** was created using different styling and layout techniques.

Module 3 – JavaScript & Interactive Development

The third module introduced JavaScript. A **To-Do List Web Application** was developed to understand user interaction, event handling, and DOM manipulation.

Module 4 – Final Project

The final module involved developing the **Smart Restaurant Booking Website** by integrating HTML, CSS, and JavaScript concepts learned throughout the internship.

The final project provided an opportunity to apply the complete web development workflow in a single project.

4.9 Expected Outcomes

The major expected outcomes of the internship were:

- Ability to create structured webpages using HTML5.
- Ability to design webpages using CSS3.
- Understanding of responsive web design principles.
- Ability to create forms and user-input interfaces.
- Understanding of JavaScript fundamentals.
- Ability to implement interactive webpage features.
- Practical knowledge of DOM manipulation and event handling.
- Ability to develop basic frontend web applications.
- Improved debugging and problem-solving skills.
- Ability to combine HTML, CSS, and JavaScript in a single project.
- Practical experience in developing responsive websites.
- Completion of multiple web development projects.
- Development of a complete **Smart Restaurant Booking Website** as the final project.

4.10 Certificates of Completion and Communication Proof

The **Internship Completion Certificate** issued by **TRL FutureX**, confirming successful completion of the **Web Development Training & Internship Program**, is attached in the relevant section of this report.



5. BIBLIOGRAPHY/REFERENCES

1. **TRL FutureX**, “Web Development Training & Internship Program,” Internship/Training Materials, 2026.
2. **TRL FutureX**, “Web Development Internship Completion Certificate,” 2026.
3. **MDN Web Docs**, “HTML: HyperText Markup Language,”
<https://developer.mozilla.org/en-US/docs/Web/HTML>
4. **MDN Web Docs**, “CSS: Cascading Style Sheets,”
<https://developer.mozilla.org/en-US/docs/Web/CSS>
5. **MDN Web Docs**, “JavaScript,”
<https://developer.mozilla.org/en-US/docs/Web/JavaScript>
6. **W3C**, “HTML and CSS Standards,”
<https://www.w3.org/standards/>
7. **JavaScript Documentation**, “JavaScript Guide,”
<https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide>
8. **Google for Developers**, “Web Development Resources,”
<https://developers.google.com/>